

FAMILY NAME:
OTHER NAME(S):
STUDENT NUMBER:
SIGNATURE:

THE UNIVERSITY OF NEW SOUTH WALES
SCHOOL OF MATHEMATICS AND STATISTICS

Exam Term 2, 2024

MATH3311/MATH5335
Computational Mathematics for Finance

- (1) TIME ALLOWED – 2 HOURS
- (2) READING TIME – 10 MINUTES
- (3) THE EXAMINATION PAPER HAS 7 PAGES
- (4) TOTAL NUMBER OF QUESTIONS TO BE ANSWERED – 4
- (5) ANSWER ALL QUESTIONS
- (6) THE QUESTIONS ARE OF EQUAL VALUE
- (7) TOTAL NUMBER OF MARKS – 80
- (8) START EACH QUESTION ON A SEPARATE BOOKLET
- (9) THIS PAPER MAY **NOT** BE RETAINED BY THE CANDIDATE
- (10) MATERIAL ALLOWED – ONE A4 PAGE WITH NOTES ON BOTH SIDES
(IT CAN BE PRINTED OR HANDWRITTEN)

All answers must be written in ink. Except where they are expressly required pencils may only be used for drawing, sketching or graphical work.

1. [20 marks]

Consider the matrix $A = [a_{ij}] \in \mathbb{R}^{n \times n}$.

- i) Assume that A has the following QR -factorization $A = Q_1 R_1$ where $Q_1 \in \mathbb{R}^{n \times n}$ is orthogonal and $R_1 \in \mathbb{R}^{n \times n}$ is upper triangular. Let $Q_2 = Q_1 S$ and $R_2 = S R_1$ where $S \in \mathbb{R}^{n \times n}$ is a diagonal matrix with entries $\in \{-1, 1\}$. Show that $A = Q_2 R_2$ is also a QR -factorization of A .
- ii) Assume now that A has a Cholesky factorisation, that is, $A = R^T R$ with $R = [r_{ij}]$ is an upper triangular matrix in $\mathbb{R}^{n \times n}$.
 - a) Show that $a_{jj} \geq (r_{ij})^2$ for $1 \leq i \leq j \leq n$.
 - b) Show that A is positive semidefinite matrix.
 - c) Assume that R is nonsingular. Then show that A positive definite.
- iii) Let $B \in \mathbb{R}^{n \times n}$ be a given non-singular upper-triangular matrix. The LU factorization with pivoting of the matrix A is: $PA = LU$.
 - a) State the properties of the matrices P , L and U .
 - b) Use the LU factorization and without computing B^{-1} , explain an efficient way of solving the linear system $B^{-1}A\mathbf{x} = \mathbf{b}$.
 - c) Estimate the computational cost of solving this linear system, given that computing the LU factorization costs $2n^3/3$ flops and computing a forward/backward substitution costs n^2 flops.
- iv) Let $C \in \mathbb{R}^{n \times n}$ be a covariance matrix, with ordered eigenvalues

$$\lambda_1 = 1.3 \times 10^{-2} < \lambda_2 = 8.1 \times 10^{-2} < \lambda_3 < \dots < \lambda_n = 5.2 \times 10^3.$$

- a) Find the condition number of C in 2-norm.
- b) Assume that the elements in C are known exactly. In the linear system $C\mathbf{x} = \mathbf{b}$, how small must the relative error in $\mathbf{b}$ be so that the relative error in computing $\mathbf{x}$ must have at least 4 significant figures?

2. [20 marks]

- i) Consider the floating point representation

$$(-1)^s(1.d_1d_2d_3)_2 \times 2^e,$$

where $s, d_1, d_3, e \in \{0, 1\}$, and $d_2 = 0$. Write down all the numbers which can be represented using this floating point representation.

- ii) Let
- f
- be a smooth function and
- $h > 0$
- . Consider the expression

$$3f(x) - 4f(x+h) + f(x+2h) = -2hf'(x) + ch^3f'''(x) + O(h^4).$$

- a) Use Taylor series expansions to find the constant c .
 b) With $h = 10^{-12}$ and $x = 10^6$, we estimated $f'(10^6)$ by computing

$$-\frac{3f(x) - 4f(x+h) + f(x+2h)}{2h}$$

using Matlab or Python (using double precision floating point operations). Justify that the expected estimation of $f'(10^6)$ is zero.

- c) Is the above approach always accurate? If not, give a counter example.

- iii) Consider the following interest rate data

i	1	2	3	4
t_i (years)	0.5	1	1.5	2
$r(t_i)$	3.3 %	3.8 %	4.0 %	3.85 %

Table 1: Interest rate $r(t)$.

In addition, it is given that $r'(0.5) = 0.011$ (the slope of r at $t = 0.5$). To estimate r over the time period $[0.5, 2]$, an analyst used the following cubic spline interpolation

$$s(t) = \begin{cases} s_1(t) = a + bt + ct^2 + dt^3, & \text{for } t \in [0.5, 1], \\ s_2(t) = \alpha + \beta t + \gamma t^2 + \delta t^3, & \text{for } t \in [1, 2]. \end{cases}$$

To compute the unknown coefficients $a, b, c, d, \alpha, \beta, \gamma, \delta$, we need 8 independent equations. Use the given data and the properties of cubic spline to state these equations. (Do not compute the unknown coefficients.)

- iv) Consider the expected value

$$E[h] = \int_1^\infty h(x)p(x) dx,$$

where $p : [1, \infty) \rightarrow \mathbb{R}$ is a probability density function (pdf) with $p(x) > 0$ for all $1 \leq x < \infty$.

- a) Define the corresponding cumulative distribution function P .
 b) Use P to transform the expected value $E[h]$ into an integral over $[0, 1]$.

3. [20 marks]

- i) Consider a real-valued random variable Y with a continuous, strictly increasing, cumulative distribution function F , so that

$$\text{Prob}(Y \leq y) = F(y).$$

- a) Show that the random variable $X = F(Y)$ is uniformly distributed in the interval $[0, 1]$.
- b) Suppose we have a procedure to generate samples $x_1, x_2, \dots$ for X . How can we use these samples to generate samples from Y ? What is likely to be the main practical difficulty with this approach?
- ii) Consider a random vector $X \in \mathbb{R}^d$ with probability density function p . The expected value of some function $f(X)$ is

$$E(f(X)) = \int_{\mathbb{R}^d} f(\mathbf{x})p(\mathbf{x})d\mathbf{x}.$$

Let $E_N(f)$ be a numerical approximation of $E(f)$.

- a) Define $E_N(f)$ using a Monte Carlo method with N samples. State the properties which the samples need to satisfy so that $E_N(f)$ can be used as an approximation of $E(f)$.
- b) What is the expected accuracy of $E_N(f)$?
- iii) With $n \geq 1$, let

$$I(g) = \int_{[-1,1]^n} g(\mathbf{x})d\mathbf{x}.$$

(Note: $[-1, 1]^n$ is an n -dimensional cubical domain.)

In the one-dimensional case (that is, when $n = 1$), we can estimate $I(g)$ using the 2-point Gauss quadrature rule as follows:

$$I(g) \approx Q_2(g) = g\left(\frac{-1}{\sqrt{3}}\right) + g\left(\frac{1}{\sqrt{3}}\right).$$

- a) Estimate $I(g)$ using a tensor product 2-point Gauss quadrature rule in the two-dimensional case, that is, when $n = 2$.
- b) How many function values do you need to calculate to estimate $I(g)$ using a tensor product 2-point Gauss quadrature rule in n dimensions?
- c) When should you avoid using such a tensor product rule to estimate $I(g)$?

iv) The stock price S is assumed to follow a geometric Brownian motion

$$\frac{dS}{S}(t) = \mu dt + \sigma(t)dW(t), \quad t \in (0, T],$$

where W is a Wiener process, μ is a positive constant, the time-dependent volatility $\sigma(t) = 0.05t$, and $S(0)$ is given.

- a) Show that $\int_0^T \sigma dW(t) = 0.05T W(T) - 0.05 \int_0^T W(t)dt$.
- b) Describe how can you simulate N asset price paths

$$(S^{(j)}(t_0), S^{(j)}(t_1), \dots, S^{(j)}(t_R)), \quad \text{for } j = 1, 2, \dots, N,$$

at times $0 = t_0 < t_1 < t_2 < \dots < t_R = T$, where $S^{(j)}(t_0) = S(0)$ is a given value. The time-points are equally spaced, i.e., with $\Delta t = T/R$, $t_n = n\Delta t$ for $n = 0, 1, 2, \dots, R$. Hint: You can use the discretizations

$$\frac{S(t_{r+1}) - S(t_r)}{S(t_r)} = \mu\Delta t + 0.05\left(t_r + \frac{\Delta t}{2}\right)\sqrt{\Delta t} Z_r,$$

where $Z_0, Z_1, \dots, Z_{R-1}$ are independent standard normal random variables.

4. [20 marks]

Recall that a European style **call** option gives the holder the right (but not the obligation) to buy the underlying asset for the strike price X at expiry T .

In this question we will use a 'call option spread' which consists of one long position (buy) of a European style call option with strike price L and one short position (sell) of a European style call option with higher strike price K (that is, $K > L$). Both options have the same expiry and the same underlying asset.

With V_K being the value of the call option in short position and V_L being the value of the call option in long position, the value $V(S, t) = V_L(S, t) - V_K(S, t)$ of the call option spread on an underlying asset with price S at time t satisfies the Black-Scholes PDE: for $0 < S < \infty$ and for $0 < t < T$,

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = rV. \quad (1)$$

We assume that the risk-free interest rate r and the volatility σ are known positive constants. We also assume that the interest compounds continuously.

- i) Draw the graph of the pay-off function of the 'call option spread', i.e. the x -axis is the price of the underlying at expiry time and the y -axis is the pay-off of the call option spread at expiry time.
- ii) As the holder of this option spread, do you benefit from an increase in the price of the underlying asset or a decrease in the price of the underlying asset?
- iii) Carefully explain what the "initial" conditions (i.e. at expiry time T) are for the call option spread.
- iv) Carefully explain what the boundary conditions are for the call option spread.
- v) Let $S_{\max} > K > L$ be the maximal price of the underlying used in the finite difference scheme and $t \in [0, T]$. Set up a grid $(S_j, t_n) = (j\Delta S, n\Delta t)$ for $j = 0, 1, 2, \dots, J+1$ and $n = 0, 1, 2, \dots, N$, where

$$\Delta S = \frac{S_{\max}}{J+1} \quad \text{and} \quad \Delta t = \frac{T}{N}.$$

Let $V_j^n \approx V(S_j, t_n)$ denote the approximate solution. We define the finite difference approximate solution V_j^n to the Black-Scholes equation (1) as:

$$\frac{V_j^{n+1} - V_j^n}{\Delta t} + rS_j \frac{V_{j+1}^n - V_{j-1}^n}{2\Delta S} + \frac{1}{2}\sigma^2 S_j^2 \frac{V_{j+1}^n - 2V_j^n + V_{j-1}^n}{(\Delta S)^2} = rV_j^n, \quad (2)$$

for $j = 1, 2, 3, \dots, J$ and $n = N-1, N-2, \dots, 1, 0$.

Rewrite the above finite difference scheme in the form

$$\alpha_j V_{j-1}^n + \beta_j V_j^n + \gamma_j V_{j+1}^n = V_j^{n+1},$$

and find the coefficients α_j , β_j and γ_j .

- vi) Use the initial condition to define the finite difference solution V_j^N for $j = 0, 1, \dots, J + 1$.
- vii) Is the scheme in (2) explicit or implicit or neither?
- viii) Let $\mathbf{V}^{(n)} = (V_1^n, V_2^n, \dots, V_J^n)^\top$. At each time level t_n , and to estimate $\mathbf{V}^{(n)}$, we have to solve a linear system of the form $A\mathbf{V}^{(n)} = \mathbf{b}^{(n)}$.
 - a) Write down the matrix A of this linear system.
 - b) Is A a dense matrix or a sparse matrix?
 - c) What is the bandwidth of A ?

END OF EXAMINATION

Q1

$$i) Q_2 R_2 = Q_1 S S R_1 = Q_1 I R_1 = Q_1 R_1 = A$$

$$* Q_2^T Q_2 = (Q_1 S)^T Q_1 S = S Q_1^T Q_1 S = S I S = S S^T = I \Rightarrow Q_2 \text{ is orthogonal}$$

$$* S = \begin{bmatrix} s_1 & & 0 \\ & \ddots & \\ 0 & & s_n \end{bmatrix}, R_1 = [r_{ij}]$$

$$S R_1 = [s_i r_{ij}], R_1 \text{ is upper triangular, so } r_{ij} = 0 \text{ for } i > j \Rightarrow s_i r_{ij} = 0 \text{ for } i > j \Rightarrow$$

$$R_2 = S R_1 \text{ is upper triangular.}$$

$$ii) A = [a_{ij}]$$

$$R = [r_{ij}], R^T = [r_{ji}], A = R^T R$$

$$a) a_{ij} = \sum_{k=1}^n r_{ki} r_{kj} \Rightarrow a_{jj} = \sum_{k=1}^n r_{kj}^2. r_{kj} = 0 \text{ for } k > j \text{ because } R \text{ is upper triangular,}$$
$$\text{so, } a_{jj} = \sum_{k=1}^j r_{kj}^2 \geq r_{jj}^2 \text{ for } 1 \leq j \leq n$$

$$b) X^T A X = X^T R^T R X = (R X)^T R X = \|R X\|_2^2 \geq 0 \text{ for any } X \in \mathbb{R}^n,$$
$$\text{so, } A \text{ is positive semi-definite.}$$

$$c) \text{ If } R X = 0 \Rightarrow X = R^{-1} 0 = 0 \Rightarrow \|R X\|_2 = 0 \Leftrightarrow X = 0$$

$$\circ \circ \text{ } \{ X^T A X = \|R X\|_2^2 > 0 \text{ whenever } X \neq 0 \Rightarrow A \text{ is positive definite.}$$

iii)

- a) L is a unitary lower triangular matrix
 U is an upper " "
 P is a permutation matrix

b) $B^{-1}AX = b \Rightarrow AX = Bb \Rightarrow PAX = PBb \Rightarrow LUX = PBb$

Let $Z_1 = PBb$, and let $Z_2 = UX$. Then, we need to solve

$$\begin{cases} LZ_2 = Z_1 \\ UX = Z_2 \end{cases}$$

c) we need about:

- n^2 flops for computing Z_1
- n^2 flops for solving $LZ_2 = Z_1$
- " " " " " $UX = Z_2$
- $2n^3/3$ flops for LU

The estimated Computational cost is $\frac{2n^3}{3} + 3n^2$

iv) a) $K(A) = \frac{\lambda_n}{\lambda_1} = \frac{5.2 \times 10^3}{1.3 \times 10^{-2}} = 4 \times 10^5$

b) $Re(x) \leq K(A) (Re(c) + Re(b)) \leq \frac{1}{2} \times 10^{-4} \Rightarrow$

$Re(c) + Re(b) \leq \frac{1}{8} \times 10^{-9}$ ~~$\Rightarrow Re(c)$~~

$Re(c) \leq \epsilon \Rightarrow Re(b) \leq \frac{10^{-9}}{8} - \epsilon = \frac{10^{-9} - 52}{8 \times 2.16} \leq 10^{-10}$

• • , $Re(b) \leq 10^{-10}$ (This is an estimation)

2) i) $s=0$, $1.d, 0d_2 \times 2^e$
 $(1.0)_2, (1.001)_2, (1.1)_2, (1.101)_2$, $e=0$

1	1.125	1.5	1.625	$e=0$
2	2.25	3	3.25	$e=1$

$s=1, -1, -1.125, -1.5, -1.625$ $e=0$
 $-2, -2.25, -3, -3.25$ $e=1$

(ii)

a) $f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + O(h^4)$
 $f(x+2h) = f(x) + 2hf'(x) + 2h^2f''(x) + \frac{4}{3}h^3f'''(x) + O(h^4)$

$$3f(x) - 4f(x+h) + f(x+2h) = -2hf'(x) + \frac{2}{3}h^3f'''(x) + O(h^4)$$

$\therefore C = \frac{2}{3}$

b) $h = 10^{-12}$, $x = 10^6$, so, $x+h = 10^6(1 + \frac{10^{-18}}{10^6}) = 10^6 = x$

~~$\frac{3f(x) - 4f(x+h) + f(x+2h)}{2h}$~~ $x+2h = 10^6(1 + \frac{2 \cdot 10^{-18}}{10^6}) = 10^6 = x$

$\therefore \frac{3f(x) - 4f(x+h) + f(x+2h)}{2h} = \frac{3f(x) - 4f(x) + f(x)}{2h} = 0$

and so, the expected estimation of $f'(10^6)$ is 0.

c) No. Example: $f(x) = x$.

$f'(10^6) = 1$ which is far away from 0.

iii)

$$\begin{aligned}
 S(0.5) = r(0.5) = 0.033 &\Rightarrow a + 0.5b + 0.5^2c + 0.5^3d = 0.033 \\
 S(1) = r(1) = 0.038 &\Rightarrow a + b + c + d = 0.038 \\
 S(1.5) = r(1.5) = 0.04 &\Rightarrow a + 1.5b + 1.5^2c + 1.5^3d = 0.04 \\
 S(2) = r(2) = 0.0385 &\Rightarrow a + 2b + 4c + 8d = 0.0385 \\
 S_1(1) = S_2(1) &\Rightarrow a + b + c + d = a + b + c + d \\
 S_1'(1) = S_2'(1) &\Rightarrow b + 2c + 3d = b + 2c + 3d \\
 S_1''(1) = S_2''(1) &\Rightarrow 2c + 6d = 2c + 6d \\
 S_1'(0.5) = r'(0.5) = 0.011 &\Rightarrow b + 2c + 3d(0.5)^2 = 0.011
 \end{aligned}$$

iv)

a) $P(x) = \int_0^x p(t) dt$

b) Let $q = P(x) \Rightarrow dq = P'(x)dx = p(x)dx$. So,

$$E[h] = \int_0^1 h(P^{-1}(q)) dq \quad \text{at } x=1, q = P(1) = 0$$

$$\text{as } x \rightarrow \infty, q = \int_1^{\infty} p(t) dt = 1$$

3) i)

$$a) P(X \leq \alpha) = P(F(Y) \leq \alpha) = P(Y \leq F^{-1}(\alpha)) = F(F^{-1}(\alpha)) = \alpha$$

So, X is uniformly distributed on $[0, 1]$.

b) $x_1, x_2, \dots$, for $X \Rightarrow F^{-1}(x_1), F^{-1}(x_2), \dots$ are samples for Y .

The difficulty is in computing the inverse of function F^{-1} .

iii)

$$a) \text{ MC method: } E_N(f) = \frac{1}{N} \sum_{i=1}^N f(x_i) \text{ where}$$

$x_1, x_2, \dots, x_N$ are N samples in $\mathbb{R}^d$ of a random variable with probability density $p(x)$

$$(E(E_N(f)) = E(f))$$

$$b) \cancel{O(N)} O(N^{-\frac{1}{2}})$$

iv)

$$a) \text{ In 2-dimensions: } Q_2(g) = g\left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right) + g\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) + g\left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right) + g\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

b) we need to evaluate 2^n functions.

c) In the case of high dimension, n is large.

iv)

a) An application of Ito's lemma implies $(\sigma(t) = 0.05t)$
deterministic

$$d(\sigma(t)W(t)) = \sigma(t)dW(t) + W(t)d\sigma(t)$$

$$\text{So, } \sigma(t)dW(t) = d(\sigma(t)W(t)) - 0.05W(t)dt$$

Integrating both sides,

$$\int_0^T \sigma(t)dW(t) = \sigma(T)W(T) - \sigma(0)W(0) - 0.05 \int_0^T W(t)dt$$

$$\int_0^T \sigma(t)dW(t) = \downarrow 0.05TW(T) - 0.05 \int_0^T W(t)dt$$

Note:

$$E\left(\int_0^T W(t)dt\right) = 0, \quad W(t) = W(t) - W(0) \sim N(0, t)$$

and so,

$$E\left(\int_0^T \sigma(t)dW(t)\right) = 0.05TW(T).$$

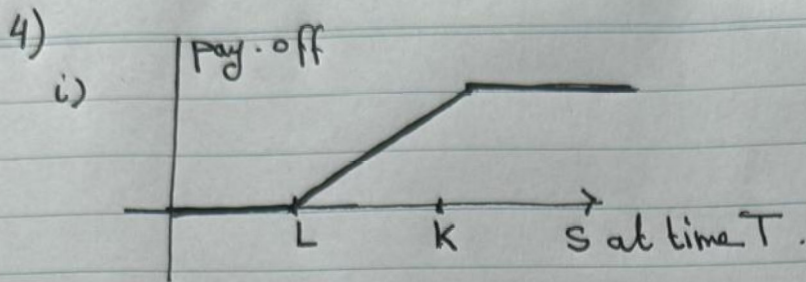
b)

$$S(t_{r+1}) = S(t_r) \left(\mu \Delta t + 0.05 \left(t_r + \frac{\Delta t}{2} \right) \sqrt{\Delta t} Z_r \right) + S(t_r)$$

Let $S^r = S(t_r)$, so,

$$S^{r+1} = S^r \left(\mu \Delta t + 1 + 0.05 \left(t_r + \frac{\Delta t}{2} \right) \sqrt{\Delta t} Z_r \right), \quad r \geq 0$$

$$\Rightarrow S^{r+1} = \underset{S(0)}{S^0} \prod_{i=0}^r \left(\mu \Delta t + 1 + 0.05 \left(t_i + \frac{\Delta t}{2} \right) \sqrt{\Delta t} Z_i \right), \quad r \geq 0.$$



ii) Increase

iii) $V(S, T) = \max(S - L, 0) - \max(S - K, 0)$

iv) $V(0, t) = 0$, $V(S, t) = (K - L)e^{-r(T-t)}$ as $S \rightarrow \infty$

v)

$$V_j^{n+1} - V_j^n + \frac{rS_j \Delta t}{2\Delta S} (V_{j+1}^n - V_{j-1}^n) + \frac{\sigma^2 S_j^2 \Delta t}{2(\Delta S)^2} (V_{j+1}^n - 2V_j^n + V_{j-1}^n) = r\Delta t V_j^n$$

$$\Rightarrow V_j^{n+1} = \underbrace{\left(r\Delta t + 1 + \frac{\sigma^2 S_j^2 \Delta t}{(\Delta S)^2} \right)}_{\beta_j} V_j^n + \underbrace{\left(\frac{rS_j \Delta t}{2\Delta S} - \frac{\sigma^2 S_j^2 \Delta t}{2(\Delta S)^2} \right)}_{\alpha_j} V_{j+1}^n$$

$$= \underbrace{\left(\frac{rS_j \Delta t}{2\Delta S} + \frac{\sigma^2 S_j^2 \Delta t}{2(\Delta S)^2} \right)}_{\gamma_j} V_{j+1}^n$$

vi) $V_j^N = V(S_j, T) = \max(j\Delta S - L, 0) - \max(j\Delta S - K, 0)$,
 $j = 0, 1, \dots, J+1$

vii) Implicit.

viii) a)

$$A = \begin{bmatrix} \beta_1 & \gamma_1 & & & \\ \alpha_2 & \beta_2 & \gamma_2 & & \\ & \ddots & \ddots & \ddots & \\ & & \alpha_J & \beta_J & \gamma_J \end{bmatrix}$$

b) A is a square matrix

c) 3.